FDA Guidance for Industry: Data Integrity and Compliance With Drug CGMP

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1. Background and Purpose

This guidance clarifies the role of data integrity in CGMP for drugs (21 CFR parts 210, 211, and 212). For the purposes of this guidance, data integrity refers to the completeness, consistency, and accuracy of data. Complete, consistent, and accurate data should be attributable, legible, contemporaneously recorded, original or a true copy, and accurate (ALCOA).

In recent years, FDA has observed CGMP violations involving data integrity during cGMP inspections. Ensuring data integrity is an important component of industry's responsibility to ensure the safety, efficacy, and quality of drugs.

Robust data governance involves organizational culture, technical controls, and routine auditing.

2. Metadata and Audit Trail Architecture

Metadata is contextual information required to understand data. An audit trail is a secure, computer-generated, time-stamped electronic record that allows for reconstruction of the course of events relating to the creation, modification, or deletion of an electronic record.

Audit trails track actions such as the modification of analytical sequences, manual integration of chromatograms, or changes to system configuration.

3. Distinction Between Static and Dynamic Records

Static record formats indicate a fixed recording (such as paper records or electronic images). Dynamic record formats allow interaction between the user and the record content (such as electronic chromatography files where integration parameters can be re-examined and manipulated).

Regulatory review requires dynamic records to be available for inspection in their native interactive format.

4. Routine Audit Trail Review Protocols

Audit trails that capture changes to critical data must be reviewed with each record and before the final approval of the record. Audit trails should be reviewed routinely by quality personnel.

Personnel conducting audit trail reviews must be adequately trained to identify unexpected alterations or anomalies.